

Pixel studio

Revise 360 lesson plan — 60 minutes

Unit	OCR J277 1.2 Memory and storage — Paper 1, Computer systems
Lesson	Pixel studio (lesson 13 of the unit)
Lesson ID	ms-l13
Experience ID	ms-l13
Plan ID	PLAN-MS-L13
Content version	159f28dd8ff1
Website	https://revise360.co.uk/experience.html?id=ms-l13

Big question, outcomes and success criteria

Big question — How does a computer store graphics?

Prerequisites

The lesson before this one is ms-l12. Pupils should already have finished it: the starter below retrieves from it.

Earlier in the unit: ms-l01, ms-l02, ms-l03, ms-l04, ms-l05, ms-l06, ms-l07, ms-l08, ms-l09-test, ms-l10, ms-l11.

Assessable outcomes

1. **ms-l13-o1** — understand how an image is a series of pixels represented in binary
2. **ms-l13-o2** — know what is meant by metadata
3. **ms-l13-o3** — understand the effect of colour depth and resolution on file size

Vocabulary, resources and preparation

Key vocabulary — Pixel, Metadata

Resources

SLIDES	presentations/MS_L13_PixelStudio_Lesson.pptx
WORKSHEET	worksheets/MS_L13_PixelStudio_Worksheet.docx
ONLINE EXPERIENCE	experience.html?id=ms-l13

Before the lesson

1. Sign in on the **teacher dashboard** (teacher.html) and open **Class logins**. Enter the class or group and the usernames, one per line, then print the login cards. A pupil cannot sign in without one, so do this before the lesson rather than in it.
2. Print the worksheet, one per pupil.
3. Open the slides and the experience once yourself, on the machine you will be teaching from, so the 360 image is already cached.
4. Check headsets or devices are charged and on the school network.

How pupils get in

1. Go to **topics.html** on revise360.co.uk.
2. Sign in with the **username and PIN from the login card**. The class is already set against the card.
3. Choose the topic, then this lesson's card.

Worth knowing before you promise anything. In this build `APP_CONFIG.backendUrl` is empty, so logins, login cards and progress are stored on the device the pupil used. The teacher dashboard says so too. A pupil who moves to a different machine starts again, so keep a class on the same devices within a unit, and treat the paper worksheet as the record that travels.

Teaching sequence

Sixty minutes including transitions. The online block is the one to move if your lesson is shorter: see the adaptation at the end.

MIN	PHASE	TEACHER	PUPILS	EVIDENCE TO CHECK
0-5	Retrieve	Starter on the board or the worksheet, from the last lesson and this lesson's prerequisites.	Answer the starter on the worksheet.	The starter box on MS_L13_PixelStudio_Worksheet.docx is filled in.
5-8	Review the starter	Take answers; correct the one most got wrong.	Correct their own starter.	A correction written, not just a tick.
8-18	Explain and model	The new idea in two or three steps, with the example worked in front of the class.	Listen and answer questions about the example.	They can say why, not only what.
18-21	Set up and settle	Sign-in, launch, and what to write before anyone taps an answer.	Sign in, open the lesson, find Pixels and bitmaps.	Everyone is in and on station 1.
21-43	Online: work the stations in order	Pupils read each wall, write the key fact and the challenge, then answer at the badge.	Read each wall, write the key fact and the challenge, then answer at the badge.	The worksheet has writing on it for each of the 6 stations before the badge is tapped.
43-48	Check understanding	The hinge question, hands down.	Answer the hinge question, hands down.	The split across the options.
48-56	Apply	Exam practice on the worksheet, in silence.	Exam practice, in silence.	worksheet, Exam practice a) to c)
56-60	Exit and set up the next lesson	The exit question and the retrieval link to next lesson.	Write the exit answer.	One sentence that answers the big question.

The explanation and the worked example

This is what the walls of the experience say, station by station. Teach the first two or three from the front before anyone puts a headset on; the rest the pupils read for themselves.

1. Pixels and bitmaps

- A bitmap image is a grid of pixels.
- Each pixel's colour is stored as a binary number.
- The computer rebuilds the image by colouring each pixel in turn.

Challenge on this wall: Colour the grid to match the image on the wall.

2. Colour depth

- Colour depth is the number of bits used for each pixel.
- 1 bit gives 2 colours; 2 bits give 4; 8 bits give 256; 24 bits give over 16 million.
- The number of colours is 2 to the power of the colour depth.

Challenge on this wall: How many colours can be stored with a colour depth of 4 bits?

3. Resolution

- Resolution is the number of pixels in the image, usually written width × height.
- More pixels means more detail, and a bigger file.
- Doubling the width and height makes the file four times bigger.

Challenge on this wall: A 400 × 300 image is doubled in both directions. How many times bigger is the file?

4. Metadata

- Metadata is data about the data: information stored with the image.
- Examples: width, height, colour depth, file type, date created, camera settings and sometimes GPS location.
- Without it, the computer wouldn't know how to display the pixels.

Challenge on this wall: State four items of metadata for a photo.

5. Calculating file size

- File size in bits = width × height × colour depth.
- Divide by 8 for bytes, then by 1,000 for kilobytes.
- Real files are a little larger because of metadata.

Challenge on this wall: Calculate the raw size of the image you coloured in, in bits and bytes.

6. Quality against size

- Increasing resolution or colour depth improves quality but increases file size.
- Choose the lowest that still looks good enough for the job.
- A website icon needs far less than a poster.

Challenge on this wall: Why might a website use a lower resolution image than a printed poster?

Retrieval and the check for understanding

Starter — Zoom right into a photo on your phone until it goes blocky. What are you actually looking at, and what is stored for each square?

Answer space: 3 lines on the worksheet.

The answer is in the teacher answers for this unit, not on this plan: a plan left on a desk should not give it away.

Hinge question

Use this one, hands down, before anyone moves on to independent work. It is a real activity from the experience (ms - l13 - s5 - c1), so the wording a pupil sees matches the wording you say.

A 200 × 100 image has a colour depth of 4 bits. What is its size in bits?

Correct: 80,000

What a pupil might choose instead

- 20,000
- 800
- 160,000

Count the hands on each. If more than a few of the class are on one wrong option, that option is the misconception to teach against before anyone goes on: go back to the station this question sits on and work through its example again, aloud, before the independent work.

The course's own response to a wrong answer here: $200 \times 100 \times 4$.

One response, not four. one response covers every wrong option; it does not say what to revisit for each one, so decide your own reteach for each option before the lesson - the list above is the start of it.

Guided to independent practice

STAGE	ACTIVITIES	WHAT IT IS
Recognise	11	The answer is on the screen and the pupil picks it: multiple choice, choosing several, matching a pair.
Construct	4	Nothing is offered. The pupil classifies, orders, fills in a table or works a number out, and types or places the answer.

Scaffolds already in the product, and how they fade

- **Hint** is a ladder, not a door: the first rung names the idea, the next shows the shape of the code, the next how it starts, the last animates the technique on different data. A pupil asks for each rung. None of them is the answer.
- **Run** is free and unmarked; **Check my answer** marks. A pupil may run as often as they like, which is what makes trial and error safe.
- **I need help** says to ask you, and says plainly that the question still has to be completed. It does not notify anyone and it does not unlock anything.
- The worked example appears on every early activity and on none of the late ones. That fade is deliberate; it is not a missing example.

Accessibility

- **Read aloud** on every task window reads the task and its brief, with the prescribed values read as values rather than as punctuation.
- Text is 16px or larger throughout, and a question window scrolls rather than shrinking its type.
- Prescribed data is boxed and in a monospace face as well as coloured, so it still reads as code in black and white, in a high-contrast mode, and to a pupil who does not separate green from amber.
- **Lower literacy and EAL:** pre-teach the vocabulary above; let the pupil say the key fact before writing it; accept the key fact in two or three words.
- **SEND:** the sequence is compulsory but the pace is not. A pupil may finish a station next lesson; the progress saves after every answer and reopens at their earliest unfinished question.
- A headset is never required. Every experience works on a laptop or a tablet with a mouse or a finger.

Delivering the online experience

Before anyone enters

- Worksheet out, pen in hand. The rule is write first, then answer at the badge: the first answer is the one that counts.
- Say where the stations are. They are numbered around the room: 1 and 2 to the right, 3 and 4 behind, 5 and 6 to the left, and the final challenge on the floor.
- Say that a badge marked **Not yet** is not a fault. It opens when the station they are on is finished.

The stations, and what to record at each

#	STATION	ACTIVITIES	RECORD ON PAPER
1	Pixels and bitmaps	2	A bitmap image is...
2	Colour depth	2	Colour depth is...
3	Resolution	2	Resolution is...
4	Metadata	2	Metadata is...
5	Calculating file size	2	Image file size is calculated by...
6	Quality against size	2	Quality and file size are linked because...
final	Final challenge: build an image	3	the answer to the station's challenge

Where to pause

Stop the class after station 2 and take answers to its challenge out loud. That is the point where a pupil who has misread the technique can still be put right cheaply.

Afterwards

Headsets off and down. Ask three pupils for the key fact from a station of your choosing, in their own words and not read off the sheet. Then the exit question.

Assessment, evidence and unfinished work

ID	QUESTION	MARKS	WHERE
ms-l13-x1	Describe what metadata is, and give two examples stored with a photo.	3	worksheet, Exam practice a); deck, Exam practice slide

ID	QUESTION	MARKS	WHERE
ms-113-x2	An image is 320 × 240 pixels with a colour depth of 4 bits. Calculate the file size in kilobytes, showing your working.	4	worksheet, Exam practice b); deck, Exam practice slide
ms-113-x3	Explain what happens to the quality and the file size when colour depth is increased.	3	worksheet, Exam practice c); deck, Exam practice slide

Expected answers

Model answers and marking guidance are teacher-only and are not printed on this plan. They are in the unit's teacher guide pack, and the per-activity marking for this lesson is in `answers/records/ms-113.json`, which is not published with the website.

Common errors

The wrong options the activities offer are the errors this lesson expects. The commonest to watch for:

- **What is a pixel?** — A type of file; A unit of storage
- **How many colours can a colour depth of 8 bits represent?** — 8; 64
- **What does resolution mean?** — The number of colours available; The size of the screen
- **An image is doubled in width and in height. What happens to the file size?** — It doubles; It stays the same
- **Why does an image file need metadata?** — To make the file bigger; To store the pixels twice

Unfinished work

Record where each pupil stopped, not a tick. They resume at their earliest unfinished question, and a station left half-done is the thing to open next lesson. Do not mark the lesson complete for a pupil who did not finish it: the score screen already shows what was and was not done, and that is the honest record.

Exit, next lesson and homework

Exit check

One of these, written, not a show of hands:

- Pixels and bitmaps
- Colour depth and resolution
- Metadata and file size

The confidence columns on the worksheet are a supplement to this, not a substitute: a pupil who rates themselves confident and cannot answer the exit question has told you nothing.

Retrieval link to next lesson

Next is `ms-114`, Sound studio. Use this lesson's big question as next lesson's starter, so the first thing they do is retrieve it.

Homework

Finish any station left open, on the website, from the question they stopped at. If the class has no devices at home, set the exam practice on the worksheet instead and mark it next lesson.

Shorter lessons, longer lessons and no internet

A 40 to 45 minute lesson

Keep the starter, the explanation and the model at full length: cutting those is what makes the online work fail. Take the time out of the online block and the apply phase. Pupils stop where they are; their progress saves after every answer and reopens at their earliest unfinished question, so the next lesson starts there.

If the internet or the devices are unavailable

Teach from the slides and the worksheet. The slides carry the same teaching as the station walls, and the worksheet carries the key facts, the challenges and the exam practice. Set the online stations when the devices are back.

There is no offline version of the experience, and this plan does not contain one. Do not hand out a printed walkthrough of the stations as a substitute: the experiences are used on the website.

